

Lab Exercise — XML: Store and Read Book Records

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Aim

Create an XML file that stores information about books, then use a programming language of your choice to read the records from the file and add new ones.

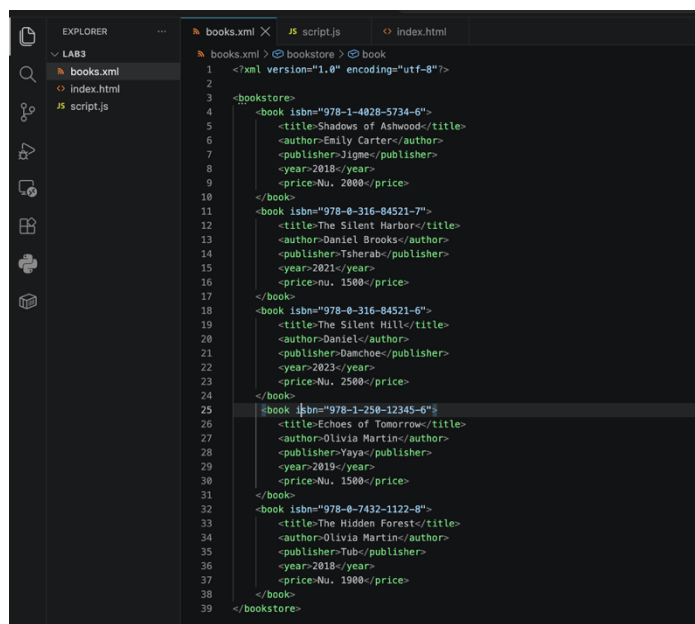
Tasks

Task 1 Create the XML file [3 marks]

Create a file named `books.xml` that stores at least **five books**. Design the structure yourself. Each book must include, at minimum:

- a unique id
- title
- author
- publisher
- year of publication • price

Your file must be well-formed. Confirm this by opening it in a web browser before you continue.



```
1 <?xml version="1.0" encoding="utf-8"?>
2
3 <bookstore>
4   <book isbn="978-1-4028-5734-6">
5     <title>Shadows of Ashwood</title>
6     <author>Emily Carter</author>
7     <publisher>Jigme</publisher>
8     <year>2018</year>
9     <price>Nu. 2000</price>
10  </book>
11  <book isbn="978-0-316-84521-7">
12    <title>The Silent Harbor</title>
13    <author>Daniel Brooks</author>
14    <publisher>Tsherab</publisher>
15    <year>2021</year>
16    <price>Nu. 1500</price>
17  </book>
18  <book isbn="978-0-316-84521-6">
19    <title>The Silent Hill</title>
20    <author>Daniel</author>
21    <publisher>Damchoe</publisher>
22    <year>2023</year>
23    <price>Nu. 2500</price>
24  </book>
25  <book isbn="978-1-250-12345-6">
26    <title>Echoes of Tomorrow</title>
27    <author>Olivia Martin</author>
28    <publisher>Yaya</publisher>
29    <year>2019</year>
30    <price>Nu. 1500</price>
31  </book>
32  <book isbn="978-0-7432-1122-8">
33    <title>The Hidden Forest</title>
34    <author>Olivia Martin</author>
35    <publisher>Tub</publisher>
36    <year>2018</year>
37    <price>Nu. 1900</price>
38  </book>
39 </bookstore>
```

Task 2 Read the file [3 marks]

Using any programming language, write a program that reads `books.xml` and displays all the book records in a readable format.

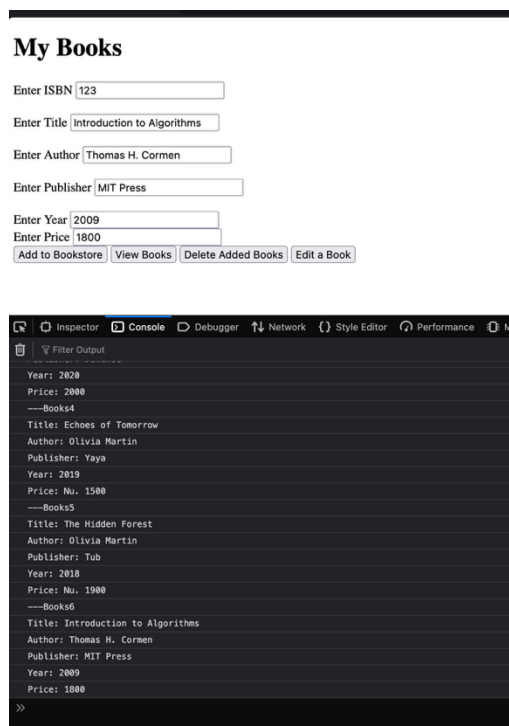
```
Price: Nu. 1900
> HTMLCollection ( 0: book , 1: book , 2: book , 3: book , 4: book , length: 5 )
---Books1
Title: Shadows of Ashwood
Author: Emily Carter
Publisher: Jigme
Year: 2018
Price: Nu. 2000
---Books2
Title: The Silent Harbor
Author: Daniel Brooks
Publisher: Tsherab
Year: 2021
Price: nu. 1500
---Books3
Title: The Silent Hill
Author: Daniel
Publisher: Danchoe
Year: 2020
Price: 2000
---Books4
Title: Echoes of Tomorrow
Author: Olivia Martin
Publisher: Yaya
Year: 2019
Price: Nu. 1500
---Books5
Title: The Hidden Forest
Author: Olivia Martin
Publisher: Tub
Year: 2018
Price: Nu. 1900
>
```

Task 3 Add a record [3 marks]

Extend your program to add the following book to `books.xml` and save the change:

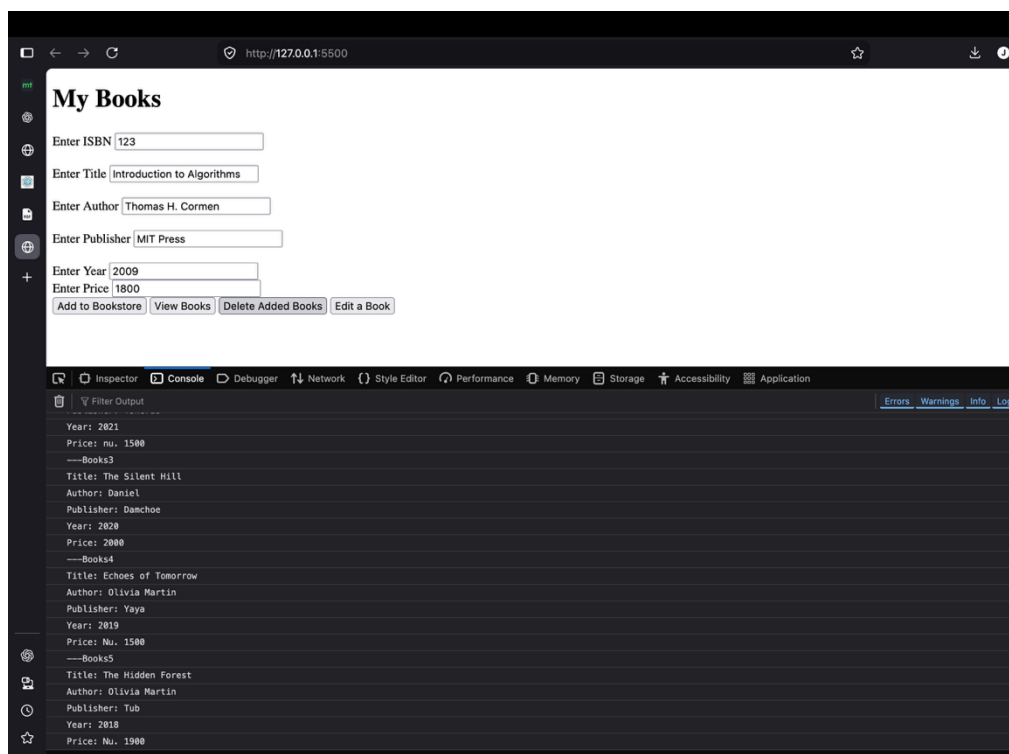
- **id:** your next unused id
- **title:** Introduction to Algorithms
- **author:** Thomas H. Cormen
- **publisher:** MIT Press
- **year:** 2009
- **price:** 1800

After adding it, read the file again to show the new book now appears.



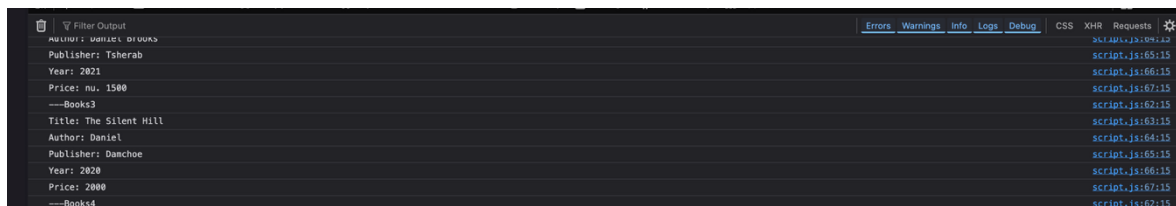
Task 4 Delete a record [3 marks]

Extend your program to delete the book you added in Task 3 (Introduction to Algorithms) from `books.xml` and save. Read the file again to show it has been removed.



Task 5 Edit a record [3 marks]

Extend your program to update one of the original five books: change its **price to 2000** and its **year to 2020**. Save the change and read the file again to show the updated values.



What to submit

- Your books.xml file.
- The source code of your program.
- The program output (paste or screenshots) showing the result of each task: books read, and the file after adding, deleting and editing a book.

Marks

This exercise is worth **15 marks** (3 marks per task) and is graded on correctness and method, the readability of your code and file, and timely submission on the VLE.

